FMGT 3130 – week 4

Q1) For 2009, Flamingo Products had a net income of $1,000,000. At 1 January 2009, there were 1,000,000 shares outstanding. On 1 July 2009, the company issued 100,000 new shares at $20 per share. The company paid $200,000 in dividends to common shareholders. What is Flamingo's basic earnings per share for 2009?

The weighted average number of shares outstanding for 2009 is

1,000,000 x (6/12) + 1,100,000 x (6/12) = 1,050,000.

Basic EPS = (Net income – preferred dividend)/(Weighted average number of shares outstanding) = ($1,000,000)/1,050,000 = $0.95

Q2) For its fiscal year-end, Sublyme Corporation reported net income of $200 million and a weighted average of 50,000,000 common shares outstanding. There are 2,000,000 convertible preferred shares outstanding that pay an annual dividend of $5. Each preferred share is convertible into two shares of the common stock. Calculate basic and diluted EPS.

Basic EPS = (Net income – preferred dividend)/(Weighted average number of shares outstanding) = ($200,000,000 – $10,000,000)/50,000,000 = $3.8

Diluted EPS = (Net income)/(Weighted average number of shares outstanding + New common shares that would have been issued at conversion)

= $200,000,000/[50,000,000 + (2,000,000 × 2)] = $3.70

Q3) For its fiscal year-end, Calvan Water Corporation (CWC) reported net income of $12 million and a weighted average of 2,000,000 common shares outstanding. The company paid $800,000 in preferred dividends and had 100,000 options outstanding with an average exercise price of $20. CWC's market price over the year averaged $25 per share. Calculate basic and diluted EPS.

Basic EPS = (Net income – preferred dividend)/(Weighted average number of shares outstanding) = ($12,000,000 – $800,000)/2,000,000 = $5.6

Diluted EPS = (Net income – Preferred dividends)/[Weighted average number of shares outstanding + (New shares that would have been issued at option exercise – Shares that could have been purchased with cash received upon exercise) × (Proportion of year during which the financial instruments were outstanding)].

The underlying assumption is that outstanding options are exercised, and then the proceeds from the issuance of new shares are used to repurchase shares already outstanding:

Proceeds from option exercise = 100,000 × $20 = $2,000,000

Shares repurchased = $2,000,000/$25 = 80,000

The net increase in shares outstanding is thus 100,000 – 80,000 = 20,000. Therefore,

the diluted EPS for CWC = ($12,000,000 – $800,000)/2,020,000 = $5.54.

Q4) Cell Services Inc. (CSI) had 1,000,000 average shares outstanding during all of 2009. During 2009, CSI also had 10,000 options outstanding with exercise prices of $10 each. The average stock price of CSI in 2009 was $15. For purposes of computing diluted earnings per share, how many shares would be used in the denominator?

With stock options, the treasury stock method must be used. Under that method, the company would receive $100,000 (10,000 × $10) and repurchase 6,667 shares ($100,000/$15). The shares for the denominator would be:

Shares outstanding	1,000,000
Options exercises	10,000
Treasury shares purchased	(6,667)
Denominator	1,003,333

Q5) Harbour Technologies reported net income of $900,000 for the year and had 300,000 weighted average common shares outstanding. The company also had convertible bonds outstanding throughout the year, incurring $120,000 in annual interest expense. These bonds can be converted into 60,000 common shares. If the company's income tax rate is 25%, calculate basic and diluted EPS.

Basic EPS = (Net income – preferred dividend)/(Weighted average number of shares outstanding) = ($900,000)/300,000 = $3

Diluted EPS = (Net income + After tax interest on convertible debt)/(Weighted average number of shares outstanding + additional common shares that would have been issued at conversion) =

($900,000 + $120,000 x (1 – 0.25)) / (300,000 + 60,000) = $2.75

Q6) Summit Manufacturing reported net income of $600,000 for the year and had 300,000 weighted average common shares outstanding. Its convertible bonds, outstanding throughout the year, incurred annual interest expense of $60,000 and can be converted into 15,000 common shares. If the company's income tax rate is 25%, calculate basic and diluted EPS.

Basic EPS = (Net income – preferred dividend)/(Weighted average number of shares outstanding) = ($600,000)/300,000 = $2

Diluted EPS = (Net income + After tax interest on convertible debt)/(Weighted average number of shares outstanding + additional common shares that would have been issued at conversion) =

($600,000 + $60,000 x (1 – 0.25)) / (300,000 + 15,000) = $2.05

Since conversion would **increase EPS** from $2.00 to $2.05, the bonds are **antidilutive** and are excluded.

Therefore, **diluted EPS remains $2.00**.

Q7) Maple Ridge Industries reported net income of $1,500,000 for the year and had 500,000 weighted average common shares outstanding. The company had 100,000 stock options outstanding throughout the year, each granting the right to purchase one common share at $15. The average market price of its common shares during the year was $25. The company also had convertible bonds outstanding throughout the year that incurred $100,000 in annual interest expense and can be converted into 50,000 common shares. Maple Ridge Industries has no preferred shares. If the company's income tax rate is 30%, calculate basic and diluted EPS.

Basic EPS = (Net income – preferred dividend)/(Weighted average number of shares outstanding) = ($1,500,000)/500,000 = $3

Assume all 100,000 options are exercised at $15 per share. The proceeds are used to repurchase shares at the average market price of $25:

Shared repurchased = 100,000 x $15 / $25 = 60,000

Incremental shares = 100,000 – 60,000 = 40,000

Diluted EPS = (Net income + After tax interest on convertible debt – preferred dividend)/(Weighted average number of shares outstanding + additional common shares that would have been issued at conversion) =

($1,500,000 + $100,000 x (1 – 0.3)) / (500,000 + 40,000 + 50,000) = $2.66